Question 2

A1 Diagram of arrangement (light source/mesh/screen or collimator/mesh/telescope) **[1]**

A2 Fringes or dots shown on screen. May be shown on diagram. **[1]**

A3 Some sensible discussion of coherence **[1]**

A4 Use of laser or single slit (and lens) in collimator **[1]**

B1 Measurements: distance from mesh to screen and separation between fringes **[1]**
OR measure an angle from the spectrometer table

B2 n = 1; find separation between central fringe and first bright fringe **[1]**
OR measure angle between central bright beam and first order beam using scale on table

B3 Use of $n\lambda = \sin\theta$ to find d. n must be clearly identified **[1]**

C Any safety precaution **[1]**
e.g. use goggles/do not look directly into laser beam/cover over sodium lamp
do not touch the bulb

D Any good/further detail **[2]**
Examples of creditworthy points might be:
Take readings with mesh in different positions to average d
Sketch/suggestion of two-dimensional array of dots on screen
Laser + mesh + screen all at same height
λ = 589 nm for sodium lamp or about λ = 630 nm for He/Ne laser/semiconductor laser
D of the order of 1 m to 4 m (laser method)
Measure 2θ and divide by two to reduce uncertainty in θ
Repeat experiment with 2nd order (3rd order etc.) beams
Detail relating to setup/use of spectrometer
Allow other valid points.

[Total: 10 marks]

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